

MINI PROJECT DOCUMENTATION

GridWatch AI

AI-Based Electricity Theft Detection System

Developed using Base44 AI

Live Project: <https://glittering-grid-guard-intel.base44.app/>

B.Tech – Electrical & Electronics Engineering

1. Abstract

Electricity theft is a major challenge for power distribution organizations because it can cause revenue loss, inaccurate demand estimation, and additional operational costs. GridWatch AI is a web-based mini-project prototype designed to identify unusual electricity-consumption patterns that may indicate possible electricity theft. The application was developed using the Base44 AI platform. It demonstrates how AI-assisted application development can be applied to a practical Electrical and Electronics Engineering problem. Flagged patterns are intended to support further human investigation and should not be treated as automatic proof of theft.

2. Introduction

Modern power distribution systems generate large amounts of electricity-consumption information. Identifying unusual behavior manually can be time-consuming, especially when the number of consumers is large. An intelligent screening system can help prioritize cases that deserve further inspection. GridWatch AI presents a web-based concept for analyzing consumption behavior and highlighting unusual patterns.

3. Problem Statement

Power utilities need efficient methods to identify consumption behavior that differs significantly from expected patterns. Manual inspection of every consumer is not practical at scale. The proposed system uses an AI-assisted web application concept to analyze consumption patterns and flag potentially suspicious cases for further review.

4. Aim

To develop a web-based AI-assisted prototype that identifies unusual electricity-consumption patterns that may indicate possible electricity theft.

5. Objectives

- Analyze electricity-consumption behavior.
- Identify unusual or abnormal patterns.
- Flag potentially suspicious consumption cases.
- Present results through a simple web interface.
- Demonstrate an AI-based solution relevant to power distribution.
- Provide a foundation for future smart-meter and IoT integration.

6. Scope of the Project

The project is an academic prototype focused on initial screening of unusual electricity-consumption behavior. It is not intended to automatically declare electricity theft. Any flagged case should be verified using technical evidence and field investigation.

7. Technology and Tools

Development Platform: Base44 AI

Application Type: Web Application

Domain: Electrical & Electronics Engineering

AI Concept: Consumption-pattern anomaly detection

Deployment: Base44-hosted web application

Documentation: GitHub

8. Proposed System

The proposed system accepts or uses electricity-consumption information, analyzes usage behavior, detects deviations from expected patterns, and displays potentially suspicious cases through a web interface. Human review remains an important final step before any enforcement decision.

9. System Workflow

Electricity Consumption Data → Data/Pattern Analysis → Unusual Behavior Detection → Suspicious Case Flagging → Web Dashboard → Human Review / Investigation

10. Working Principle

1. Electricity-consumption information is considered as input.
2. Consumption behavior is analyzed.
3. The application identifies unusual deviations or patterns.
4. Potentially suspicious cases are flagged.
5. Results are presented through the web interface.
6. Authorized personnel can prioritize flagged cases for further investigation.

11. Key Features

- Web-based monitoring interface.
- Consumption-pattern analysis concept.
- Identification of unusual usage behavior.
- Dashboard-oriented presentation.
- AI-assisted application development using Base44.
- Responsive prototype suitable for demonstration.

12. Advantages

- Supports faster initial identification of unusual consumption.
- Can help prioritize field inspections.
- Reduces manual effort during initial screening.
- Provides a practical AI application for an EEE problem.
- Can be extended to smart-grid and IoT environments.

13. Limitations

- A suspicious pattern does not prove electricity theft.
- Detection quality depends on the available data.
- Real-world deployment requires secure utility/customer data.
- False positives and false negatives are possible.

- Production deployment requires validation with historical utility data and domain experts.

14. Future Scope

Future versions can integrate smart meters and IoT devices, real-time monitoring, machine-learning anomaly detection, customer risk scoring, GIS-based visualization, automated alerts, historical trend analysis, role-based authentication, secure data storage, and model-performance evaluation using real labeled utility data.

15. Testing

The prototype can be evaluated using normal consumption examples, sudden consumption changes, repeated unusual patterns, missing/incomplete data, and different customer/load profiles. The application should also be tested for responsive behavior on desktop and mobile browsers.

16. Expected Result

The expected result is a web application that helps users identify electricity-consumption patterns that appear unusual and highlights those cases for further review and investigation.

17. Conclusion

GridWatch AI demonstrates how AI-assisted web development can be applied to a practical power-distribution problem. By analyzing electricity-consumption behavior and flagging unusual patterns, the project provides a starting point for intelligent electricity-theft screening. The prototype can be enhanced into a more advanced smart-grid solution through real-time smart-meter integration, machine-learning models, IoT devices, GIS mapping, and secure utility systems.

18. GitHub and Project Link

The project documentation can be maintained in a GitHub repository. The live Base44 application is available at:

<https://glittering-grid-guard-intel.base44.app/>

Recommended repository name: **AI-Electricity-Theft-Detection-GridWatch**

19. References

- Base44-hosted project: <https://glittering-grid-guard-intel.base44.app/>
- GitHub Documentation: <https://docs.github.com/>
- General concepts of anomaly detection and smart-grid monitoring.